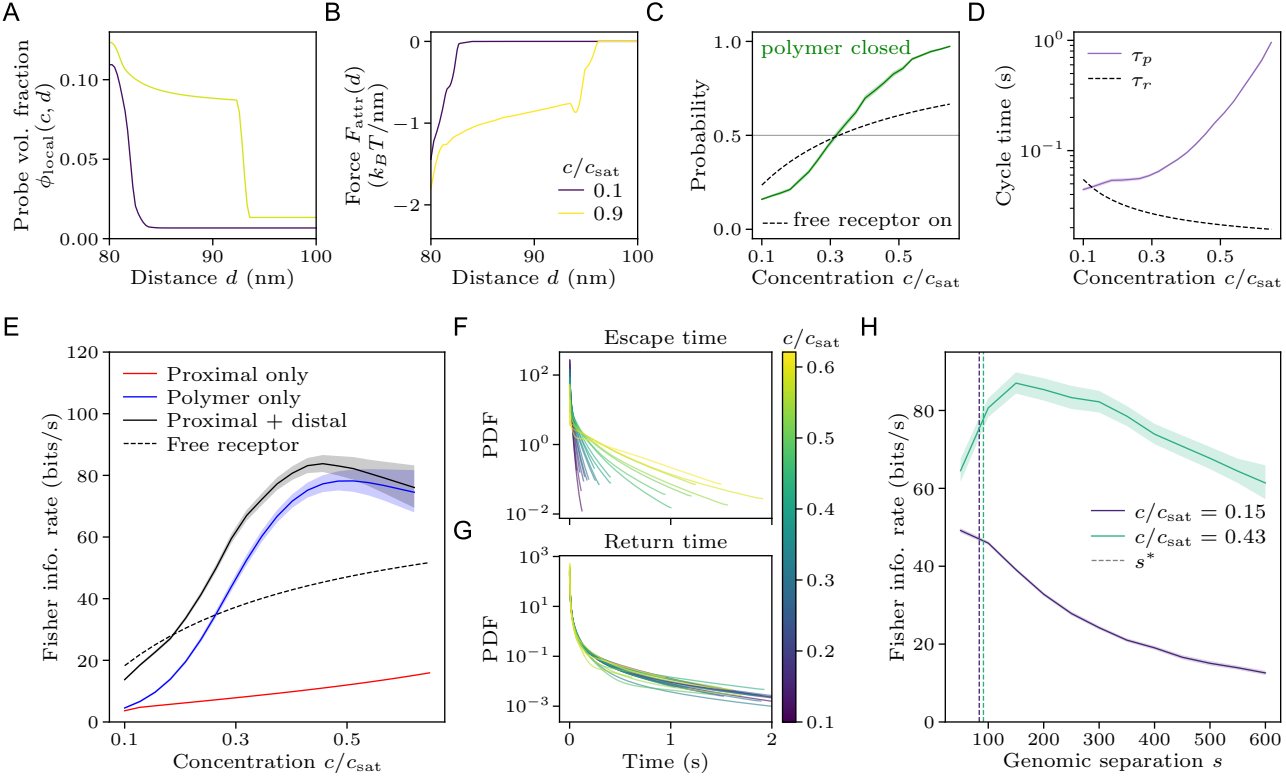

A
Probe vol. fraction $\phi_{\mathrm{local}}(c,d)$
Distance d (nm)
B
Force $F_{\mathrm{attr}}(d)$ (k_BT/nm)
c/c_{sat}
0.1
0.9
Distance d (nm)
C
polymer closed
free receptor on
Probability
Concentration c/c_{sat}
D
τ_p
τ_r
Cycle time (s)
Concentration c/c_{sat}
E
Proximal only
Polymer only
Proximal + distal
Free receptor
Fisher info. rate (bits/s)
Concentration c/c_{sat}
F
Escape time
PDF
G
Return time
PDF
Time (s)
c/c_{sat}
H
$c/c_{\mathrm{sat}} = 0.15$
$c/c_{\mathrm{sat}} = 0.43$
s^*
Fisher info. rate (bits/s)
Genomic separation s